

Return-to-Play, Return-to-Learn, Return-to-Work Ladder

Staged graduated progression · Amsterdam 2023 aligned

Concussion recovery uses a staged graduated progression. Each stage introduces controlled increase in demand. Progression requires **at least 24 hours** at each stage without new or worsening symptoms. Symptom exacerbation = drop back one stage for 24 hours, then re-attempt.

Medical clearance is required before a patient returns to contact practice (stage 5) and full sport (stage 6). Do not clear prematurely under coach, parent, or patient pressure. Document your reasoning.

Return to Sport — 6-stage ladder

Stage	Activity	Examples	Progression criteria
1	Symptom-limited activity	Walking around house; light ADLs; very light reading; brief screen time as tolerated	No worsening of symptoms with baseline activities
2	Light aerobic exercise	Walking (outdoors), stationary cycling at low intensity, < 70% max HR, 15–20 min	Tolerated without new or worse symptoms in the next 24 hrs
3	Sport-specific exercise	Running drills, skating, swimming drills; no head impact, no rotational loading	Tolerated; adds movement complexity
4	Non-contact training drills	Passing drills, ball handling, skating drills with team; progressive resistance exercise	Tolerated; full exercise + coordination + cognitive load
5	Full contact practice	Normal training session including contact — requires medical clearance first	Tolerated; restore confidence and assess by coaching staff
6	Return to sport	Normal game play	No restrictions

Return to Learn (school) — 5-stage ladder

Stage	Activity	School accommodations
1	Daily activities at home, no schoolwork	Medical certificate; no screens/reading if provocative

Stage	Activity	School accommodations
2	School activities at home	15–30 min blocks of reading/homework with breaks; no tests
3	Return to school part-time	Half days initially; quiet room available; extended deadlines; no PE; reduced screen time; modified timetable prioritising core subjects
4	Return to school full-time with modifications	Full days; continued extra breaks and deadline extensions as needed; graduated PE re-introduction (non-contact)
5	Full return, no modifications	Normal school participation including contact sport if cleared

Return to Learn before Return to Sport. A child who cannot manage full school days should not be returning to contact sport. This is a common point of disagreement with parents and coaches — hold the line, and document the reasoning.

Return to Work — 5-stage ladder

Stage	Activity	Workplace modifications
1	Daily activities at home, no work tasks	Medical certificate; focus on rest + rehab
2	Light work activities from home	Administrative, reading, email in short blocks; < 2 hrs/day
3	Return to work part-time, modified	Half-days initially; quiet workspace; reduced screen time; scheduled breaks; no safety-critical tasks
4	Return to work full-time, modified	Full days; continued modifications as needed; graduated re-introduction of complex tasks
5	Full return, no modifications	Normal duties including safety-critical activities if cleared

Safety-critical occupations — driving, machinery operation, emergency services, defence, elite sport — require explicit clearance prior to return to those duties. Document clearance with the specific tasks specified.

Return to Driving

Frequently overlooked. Concussion produces slowed reaction time, reduced attention, and impaired visual processing — all impair driving safety.

Do not drive while symptomatic. Clearance requires:

- Absence of symptoms at rest
- Absence of symptoms during cognitive demand (reading, screen)
- Normal visual function (acuity, fields, convergence, pursuit/saccades)

- Normal cognitive screen
- Patient's subjective sense of readiness

For uncertainty, refer to an OT driving assessment.

BCTT-guided exercise prescription

For stages 2 onwards, the Buffalo Concussion Treadmill Test quantifies the symptom-provocation threshold.

Prescribe aerobic exercise at 80–90% of threshold heart rate.

BCTT protocol summary:

- Start at 3.2 mph (5.2 km/h), 0% grade, walking
- Increase grade by 1% each minute up to 15%
- Then increase speed by 0.4 mph each minute
- Symptom rating pre-test and every minute
- Test stops at: symptom exacerbation ≥ 3 points from baseline, RPE > 17 , or 20 minutes
- HR at symptom exacerbation = **symptom threshold HR**
- Prescribe training at 80–90% of threshold HR, 5–6 days/week, progressing as tolerated

Alternative: **MOVE Protocol** for telehealth — 7 bodyweight stages $\times$ 60 seconds, equivalent feasibility and safety data (Teel 2023).

Clearance documentation template

A return-to-play / work / school clearance note should include:

1. **Patient identifiers:** name, DOB, date of clearance
2. **Injury details:** mechanism, date of injury, initial clinical findings
3. **Recovery trajectory:** SCOAT6 progression, examination findings
4. **Progression completed:** specific stages completed, dates, any symptom exacerbations
5. **Current examination:** SCOAT6 symptom score (should be at or near baseline), VOMS normal or improved, balance normal, cognitive screen normal
6. **Specific clearance:** exactly what the patient is cleared for (stage 6 return to contact rugby; return to full-time work including driving; etc.)
7. **Any remaining restrictions or cautions**
8. **Clinician signature, AHPRA registration number, date**

Special considerations

Adolescents. School precedes sport. Slower recovery. Parental involvement. Child SCAT6/SCOAT6 for < 13 .

Professional athletes. Governing body protocols typically override general consensus and may be more or less conservative than this ladder.

Military and emergency services. Specialised clearance pathways exist; refer to Defence or agency-specific frameworks.

Repeated concussions. Lower threshold for conservative clearance; longer observation at each stage; consider specialist input for return-to-contact decisions.

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Aligned with the Amsterdam Consensus Statement on Concussion in Sport (Patricios et al. 2023), the AIS/SMA Concussion and Brain Health Position Statement (2024), and AHPRA Code of Conduct. For qualified allied health practitioners. Not a substitute for independent clinical judgement.